

CLACS Binder

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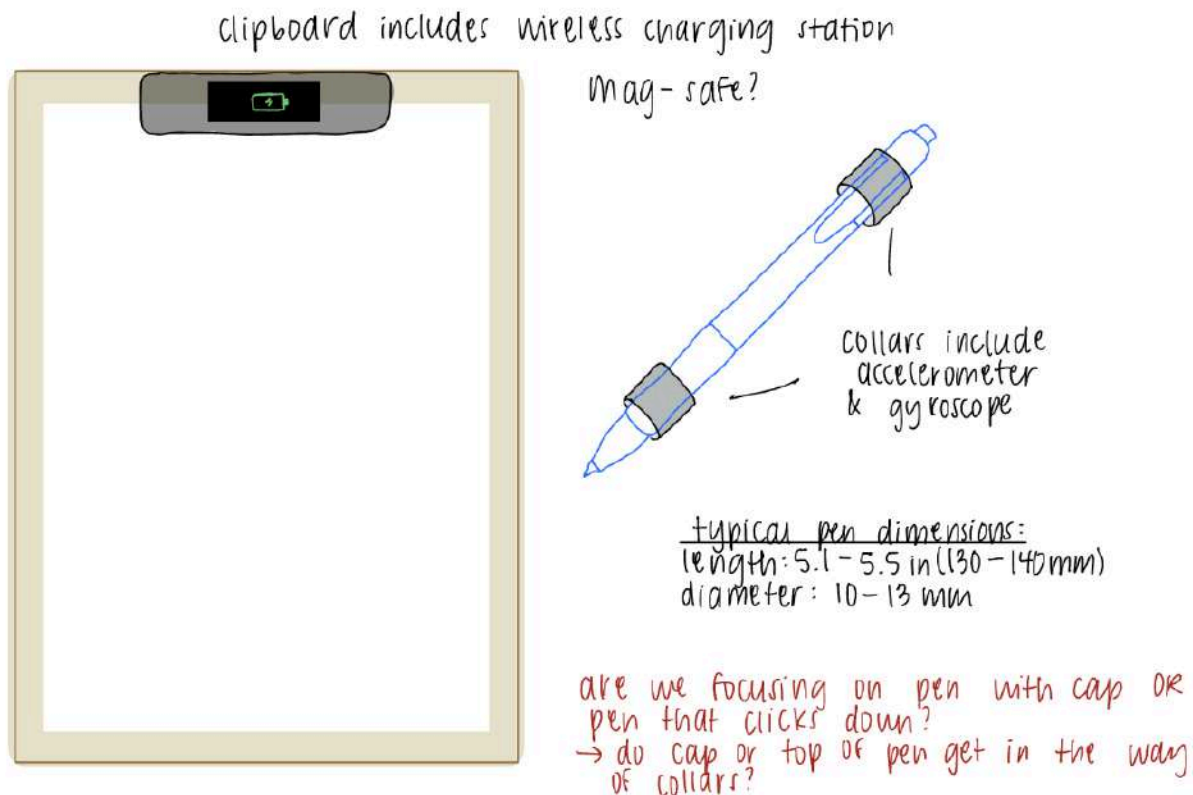
Design Diary:

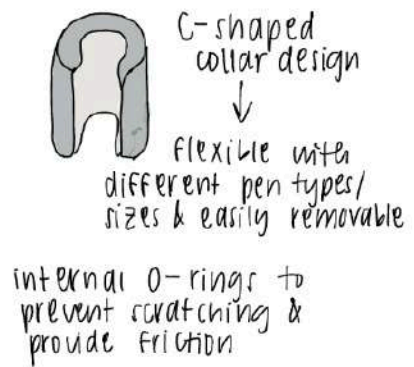
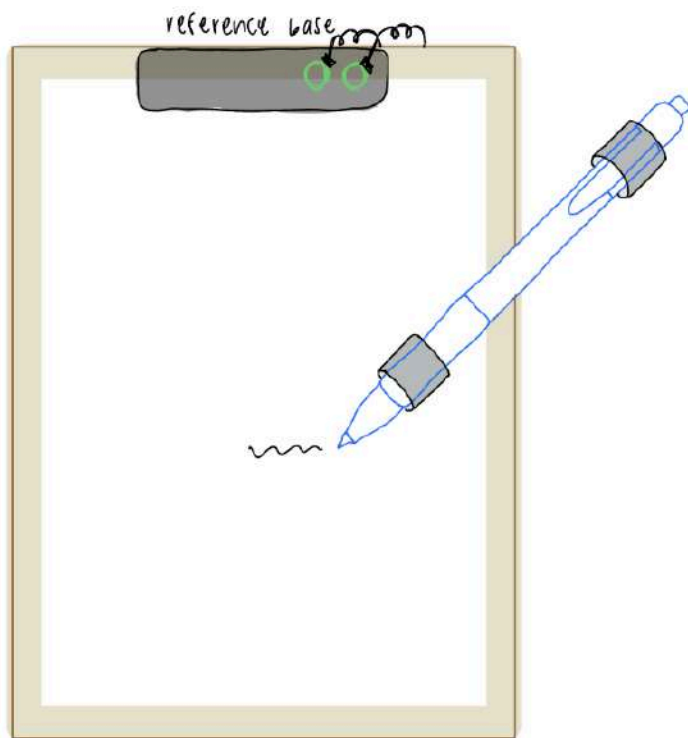
--- 2025-09-26 --- FIRST TEAM MEETING ---

All five team members met for the first time as a group. The first draft of the project proposal was submitted. The team discussed how to split up the project into subprojects (Circuit & PCB, Battery & Charging, Signal Processing & Algorithm, App & UI). Weekly meetings were set for Tuesdays between 11am and 2pm. The team agreed each member would submit 5 potential project names and a drawing of their vision for the pen attachment before the next meeting.

Action items: Set up GitHub, project name submissions, vision drawings.

The images below show some of the first vision drawings for this project based on our first few brainstorming sessions:





Project Timeline:

Sep 15, 2025 — Group chat created. Initial scheduling discussions to find a common meeting time.

Sep 17, 2025 — First in-person team meeting at Carlton Commons. Project ideas discussed.

Sep 24, 2025 — Chris joins the team! The team discusses scheduling recurring meetings with Professor Vallancourt, settling on Tuesday/Thursday availability.

Sep 25–26, 2025 — Group meets at Uris Library. Project structure discussed: agreement to divide work into 3–5 clear subprojects with each person owning 1–2. GitHub repository created and scaffolded out. Sekander uploads an early pen attachment sketch from the team meeting.

Oct 3, 2025 — Slide deck for faculty presentation created and shared. The team discusses needing drawings and visuals for the project description slide.

Oct 7, 2025 — First formal meeting with Professor Vallancourt in person (room 1332, Mudd). Key decisions made:

- Concept: Professor Vallancourt suggests using a clipboard as both the writing surface and the positioning reference for the pen.
- Budget: \$500/team standard; up to \$1,000 acceptable with justification. Individual purchases with reimbursement preferred over slow department ordering.
- Sensors: Team directed to verify sensors against datasheets from the rough draft proposal. Flagged as harder than typical prior teams who used standalone boards without dealing with raw sensor data.

- Positioning algorithm: Identified as one of the hardest parts of the project; cannot begin until sensor selection is finalized.
- Data transfer: Professor Vallancourt recommends a WHOOP-style approach, store data locally on the device and transfer in batches. Better battery life, allows cordless use.
- Battery: Team directed to speak with Professor Preindl.

Oct 12, 2025 — Team meeting at Butler Library. Topics:

- Weekly meeting time slot confirmed: Tuesdays 11am–1pm.
- Schedule meeting with Professor Preindl about battery design.
- IMU sensor selection is still to be determined.
- KiCad chosen for PCB design; Python chosen as the programming language.
- LIDAR is raised as a fallback if IMU proves non-viable. Decision: develop both IMU and LIDAR paths in parallel and compare.

Oct 14, 2025 — Professor Vallancourt meeting. New directions:

- Team directed to explore resistive touchscreen technology as an additional alternative positioning approach.
- Decision to purchase an IMU sensor board for viability testing.

Oct 17, 2025 — IMU proof-of-concept plan documented. Hardware ordered:

- Adafruit IMU adapter board (\$16.20)
- ST evaluation board (STEVAL-MKI109V3) (\$106.00)

Oct 20–21, 2025 — Slides prepared for the faculty proposal presentation. Block diagram created and reviewed.

Oct 21, 2025 — Professor Vallancourt meeting. Magnetic sensors were raised as another alternative positioning approach alongside IMU and LIDAR.

Oct 24, 2025 — Faculty presentation delivered.

Nov 11, 2025 — Professor Vallancourt meeting. The team notes that parts are still pending; real data collection and progress cannot begin until they arrive.

Nov 18, 2025 — Professor Vallancourt meeting. The parts package arrives and is picked up the same day.

Nov 21, 2025 — Team meets to examine the parts.

Dec 8, 2025 — Project constraints document completed and submitted.

Jan 27–28, 2026 — The ST evaluation boards with the LSM6DSO16IS IMU are successfully connected to a computer and recognized for the first time.

- Mac used; MEMS Studio software worked without additional configuration.
- Firmware updated via STM32CubeProgrammer.
- Windows compatibility issues noted; two separate Windows PCs failed over break, Mac confirmed working.
- Materials stored in room 1206 going forward.

Jan 30, 2026 — Second milestone: raw IMU data successfully collected from the test sensor.

- Sensor confirmed working on Albert's laptop (also Mac).
- Basic accelerometer samples recorded.
- Discussion of a 3D-printed enclosure for cleaner testing.
- First scripts written to load and inspect the IMU data output and attempt a basic trajectory reconstruction via double integration of acceleration.
- Early ML pipeline stub started for letter recognition.

Feb 3, 2026 — Professor Vallancourt meeting. Topics:

- IMU is working; initial data gathered.
- Next week's meeting with Professor Vallancourt will be held in room 1206 to demonstrate the working sensor live.
- Goal for the week: more letter data and improved code output.
- Double-integration method assessed as not working well; CNN approach proposed, though it requires significantly more data.

First in-lab letter writing test results recorded and saved.

Feb 5, 2026 — Data collection protocol established:

- 5 samples per letter minimum
- 104 Hz sampling rate
- Both accelerometer and gyroscope recorded
- Lowercase letters as the target
- One page takes roughly 5 minutes to complete

Feb 6, 2026 — Large data collection day. Multiple team members contribute sensor recordings covering several letters. First trained models produced. Initial accuracy around 10%, attributed to small sample size. Models are basic CNNs on raw accelerometer data; low accuracy was expected at this stage.

Feb 8, 2026 — ML approach revised. Rather than classifying directly from raw IMU sequences, the pipeline is changed to map IMU data to a 64x64 image representing the shape of the letter, using traced handwriting samples as ground truth. 55+ individual handwriting samples for the letter 'a' collected and averaged.

Proof-of-viability declared complete:

- Data preparation pipeline done
- LSTM+CNN sequence-to-image model built, trained on ~12 pages = 1680 individual samples, producing recognizable letter 'a' outputs
- Optimal classification threshold identified

Feb 10, 2026 — Team decides to pivot from KiCad to Altium Designer for schematic and PCB design. All team members sign up for Altium Designer student licenses.

Feb 17, 2026 — Prints picked up from the makerspace. Physical hardware fit test conducted and pen attachment component is confirmed to seat fully.

Feb 18–19, 2026 — More IMU data collected. Data collection parameters confirmed: 2-second interval per box. Physical test setup photographed and documented.

Feb 21, 2026 — Overfitting problem formally identified. With only ~12 samples per letter the model memorizes the training set and cannot generalize. More data needed before further model development is meaningful.

Feb 22, 2026 — New model iteration attempted to address overfitting. Additional training data collected.

Feb 26–27, 2026 — New package of IMUs arrives. Picked up and stored in 1206.

Mar 1, 2026 — Another round of CNN training with the expanded dataset. Accuracy and generalization are still being worked through.

Mar 3, 2026 — Professor Vallancourt meeting. Proposal for STM32WB55 as the MCU, citing its BLE capability and well-documented ST firmware interface.

Mar 5–7, 2026 — Additional data collection from multiple team members. The dataset is expanded with more per-letter samples.

Mar 8, 2026 — Schematic updated: connections between MCU, IMU, and antenna revised.

Mar 10, 2026 — Group meeting to finish the schematic and begin PCB layout in Altium. PCB task list established:

- Choose and order battery (~100mAh, 2.5x2cm, compatible with charger IC)
- Select correct charging resistor
- Finalize remaining BOM components

Mar 14, 2026 — Battery candidate identified: Adafruit 4237 LiPo, shared for team review.

Mar 23–24, 2026 — Block diagram from the original proposal used to guide the interim report draft. Schematic and PCB layout progress. PCB still needs switch placement and copper plane assignments resolved.

Mar 26, 2026 — PCB changes nearly complete; remaining DRC errors reviewed and non-relevant ones waived.

Mar 27–28, 2026 — PCB changes finalized. Goal set: finalize BOM, submit electrical review, and send PCB files to fabrication before the next Professor Vallancourt meeting.

Mar 30, 2026 — BOM consolidated and cleaned up. All parts found in stock on DigiKey; a few resistors substituted for equivalent values and footprints. DigiKey cart assembled.

Mar 31, 2026 — PCB files submitted for fabrication.

Apr 1, 2026 — LiPo battery added to BOM. DigiKey parts order submitted.

Apr 8, 2026 — Abstract drafted, reviewed by the group, and submitted.

Apr 13, 2026 — Ordered parts arrive. Software task checklist created and shared.

Apr 14, 2026 — Team meets to inspect received parts and confirm everything is present. Parts inventory verified. Confirmed ready to begin assembly.

Apr 17, 2026 — Major assembly session in room 1206. MCU side was fabricated using the oven in 1233. Power side will be hand soldered. Poster work started. Housing and clip-on attachment design in progress. Connection issues identified between MCU and computer, debugging begins.

Apr 18, 2026 — MCU pin shorting cleaned up. Poster updated with additional content.

Apr 21, 2026 — Professor Vallancourt zoom meeting; PCB shown to him from room 1206. Poster and one-slide PowerPoint submitted. The team begins reaching out to professional soldering companies for potential assembly assistance.

Apr 23, 2026 — PCB continuity check performed: 3.3V and GND confirmed not shorted, SMD components not shorted. PCB declared ready for initial power-on. Connection issues persist between MCU and computer still present. PCB assembly 2nd try on a new board, the team decides to try using the oven for both sides of the board. SMD reflow completed using the oven (both sides)

Apr 24, 2026 — Work session in room 1233. PCB design flaw discovered from the fabricated board: USB-C data pins routed to the wrong MCU pins. Three remediation options identified and discussed with Professor Vallancourt. MCU/IMU pinout documented.

Apr 25, 2026 (overnight) — USB-C wiring flaw fully diagnosed. The required fix involves physically rerouting three adjacent MCU pins. Pulling BOOT0 high confirmed not sufficient as a workaround. Schematic corrected with all fixes, plus additions: charging LED fixed, BLE status indicator LED added, 3.3V rail indicator LED added, breakout header added for the affected pins.

Apr 25, 2026 — Professor Vallancourt meeting. USB-C breakout board built from a spare PCB as a jumper-wire workaround for the current board. Software confirmed working end-to-end. BLE connection not yet established. PCB V2 ordered. STM32WB55 boot mode documentation reviewed.

Apr 26, 2026 — Parts order submitted for PCB V2 additional components. Clip-on housing design nearly complete, not yet printed. USB-C power-only cable sourced for testing.

Apr 27, 2026 — Pin rework tested and ST-Link connection to MCU established at 1800 Hz.

Apr 29, 2026 — PCB V2 fabrication complete and shipped.

Apr 30, 2026 — PCB V2 estimated to arrive Tuesday (May 5); component parts expected Monday. Plan: use remaining time to develop and test software on the current board; assemble and flash V2 upon arrival (~6 hours estimated). Firmware development for the STM32WB55 begins.

May 1, 2026 — Extended I2C diagnostic firmware written with progressive test phases, from basic clock checks through raw SDA/SCL line monitoring, to isolate the communication failure between the MCU and IMU.

Root cause of IMU non-function confirmed: the SCL and SDA lines are physically bridged to an adjacent ground pad by solder on the fabricated board, preventing I2C communication entirely. The same layout flaw is present on the incoming V2 board. Workarounds discussed: a 100um laser or fine X-acto knife cut on the affected trace isolates the pins without damaging the board.

Real-time handwriting demo application built from scratch. The app reads live IMU data over USB or BLE, detects pen strokes, runs the letter recognition model, and renders predicted letters on a scrollable canvas with PDF export. Real-time connection between the test board and the app confirmed working.

May 2, 2026 — First successfully exported PDF of a full handwriting session saved as a reference.

May 3, 2026 — Exacto knife cut confirmed viable on a spare board (no laser cutter needed). Casing update expected from Andres the following day. The team plans to use the existing test board for Bluetooth integration work while waiting for V2.

May 4, 2026 — Stroke detection error identified. Writing ~55 strokes was producing only 2–12 detected strokes. The fixed amplitude threshold could not work because hand tremor and pen repositioning between letters kept the signal continuously above any fixed cutoff, merging letters together.

The detection signal changed from raw acceleration magnitude to frame-to-frame acceleration delta, which cancels gravity automatically. Threshold is made adaptive, ending a stroke when the signal drops below 50% of its recent rolling peak rather than a fixed floor. Self-normalizes across different writing strengths with no per-user calibration. Result: 57 strokes detected against 55 expected (previous best: 17).

Additional improvements: model inference moved off the UI thread to prevent freezing during prediction. Time normalization improved to better preserve letter shape across different writing speeds.

PCB V2 and stencil arrive and placed in room 1206.

PCB V2 board 1 fabricated and tested. Can't connect MCU to computer, issue identified as damaged USB-C header and possible physical MCU pin connection errors.

Missing components for a 2nd try ordered on Mouser with priority overnight shipping (expected Wednesday morning).

May 5, 2026 — Transfer learning and letter classification integrated into the app. The model is pretrained on a public online handwriting dataset then fine-tuned on the team's own IMU data. Interactive labeling and label review tools built to manage the training dataset. Letter classifier integrated into the live app: the canvas now shows top predicted letters below each recognized stroke. Final evaluation dataset collected by Caitlin. Professor Mouser overnight parts were delayed several hours due to van mechanical issues. Additional training data collected from recruited participants.

May 6, 2026 — More training data collected; additional participants recruited. PCB V2 board number 2 fabrication completed. The board confirmed working for data transfer from IMU to MCU to computer.

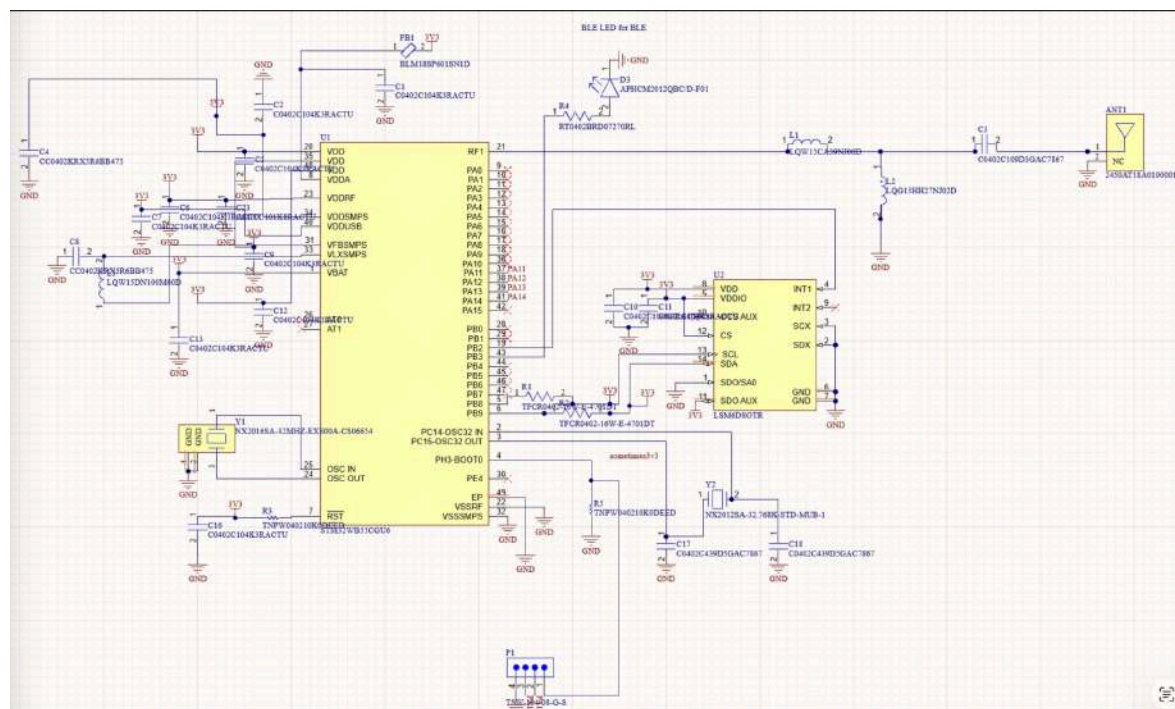
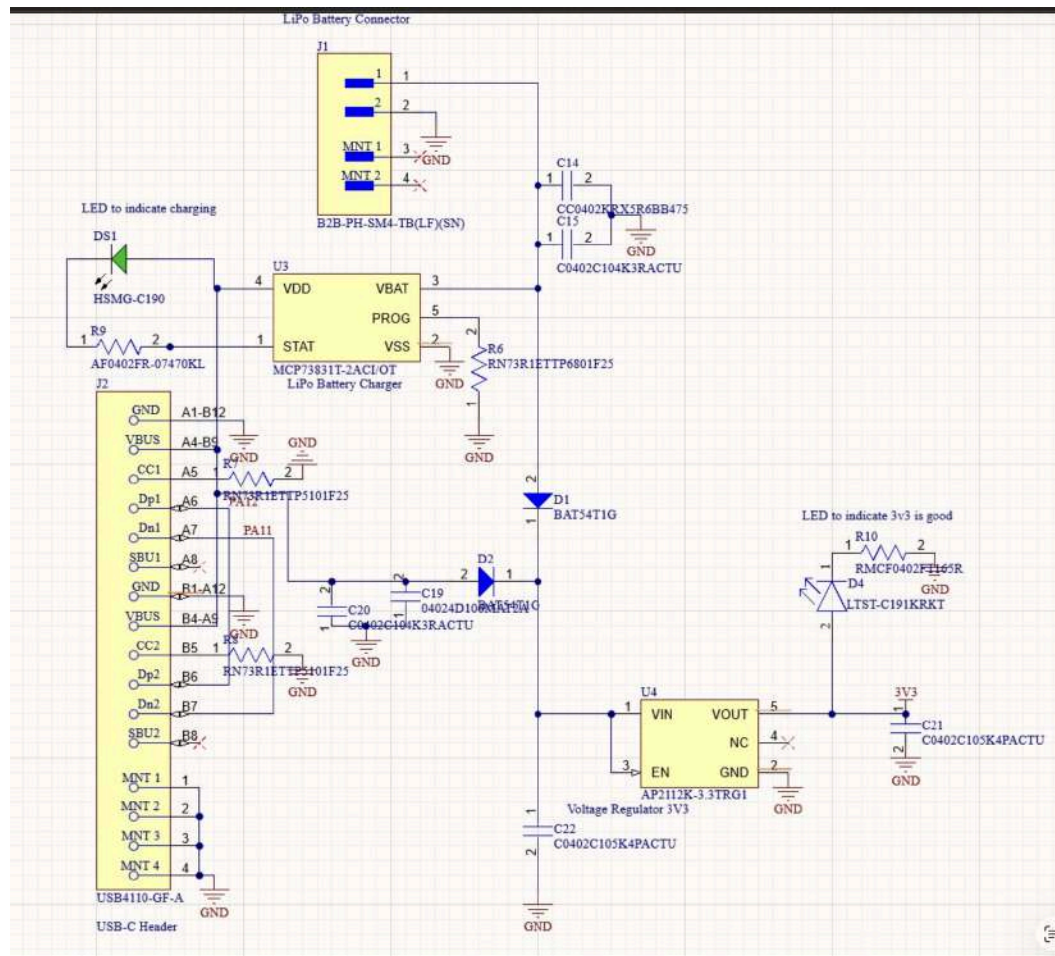
May 7, 2026 — Finalized board discovered to not work with already collected dataset. Small sample set collected on new hardware, new PNN trained on this data to recognize which letter was written. Expo day. CLACS presented at the Senior Design Expo.

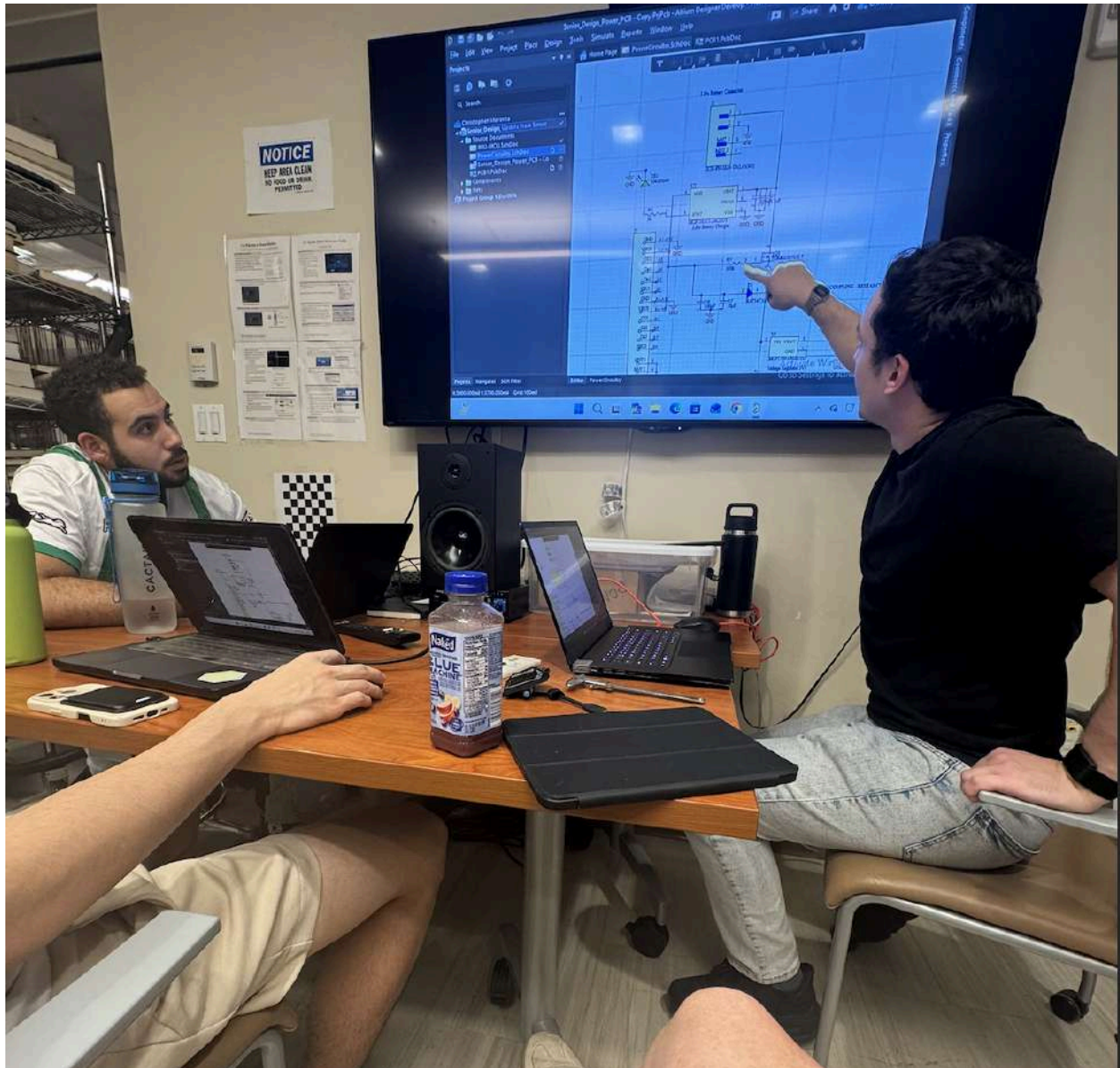
Case 2

a	b	c	d	e	f	g	h	i	j
k	l	m	n	o	p	q	r	s	t
u	v	w	x	y	z	a	b	c	d
e	f	g	h	i	c	z	a	r	s
t	u	v	w	x	y	z	a	b	c
d	e	f	g	h	i	j	k	l	m
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h	f	x	o	p	b	c	d	e	f
n	o	p	q	r	s	t	d	r	j
a	b	c	d	e	f	g	k	l	m
n	o	p	q	r	s	t	u	v	w

The image above shows an example of what one of the image uploads looked like. Each group member spent time collecting multiple pages of data in order to train the model.

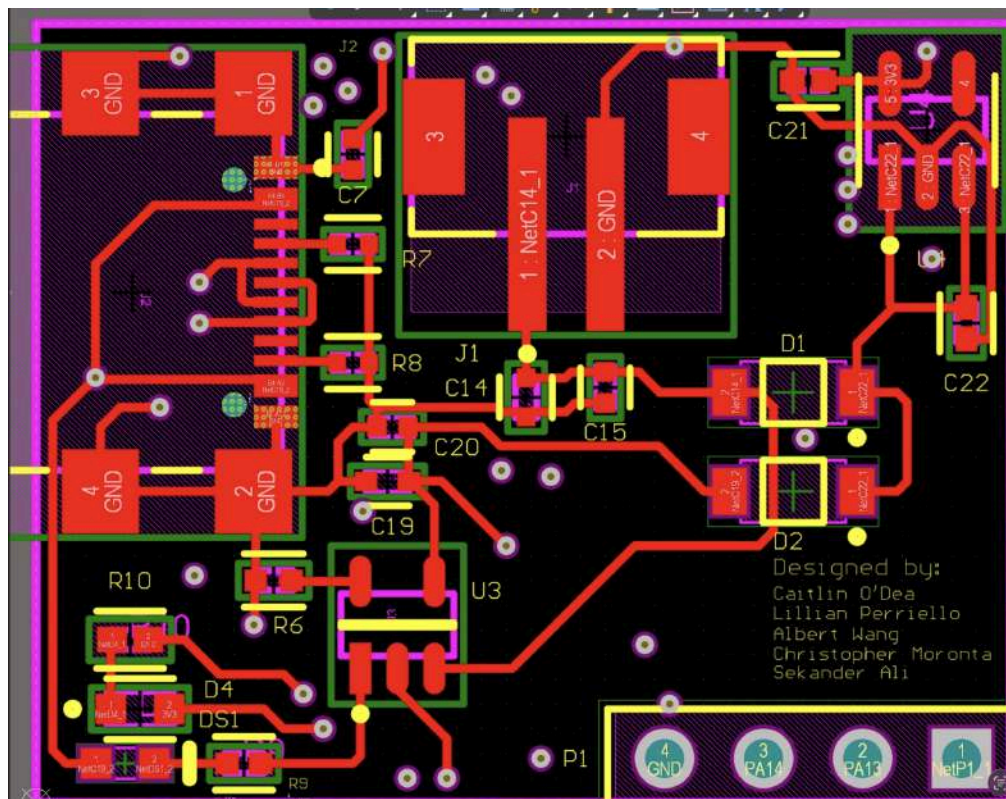
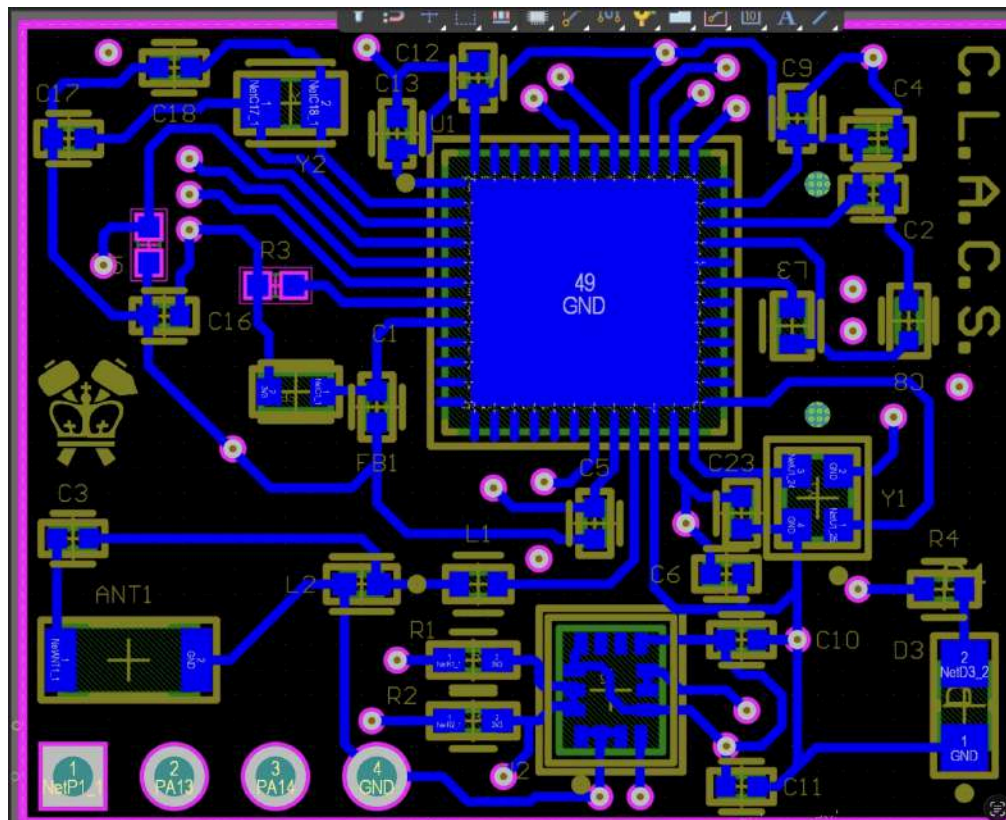
Images showing the full design process:

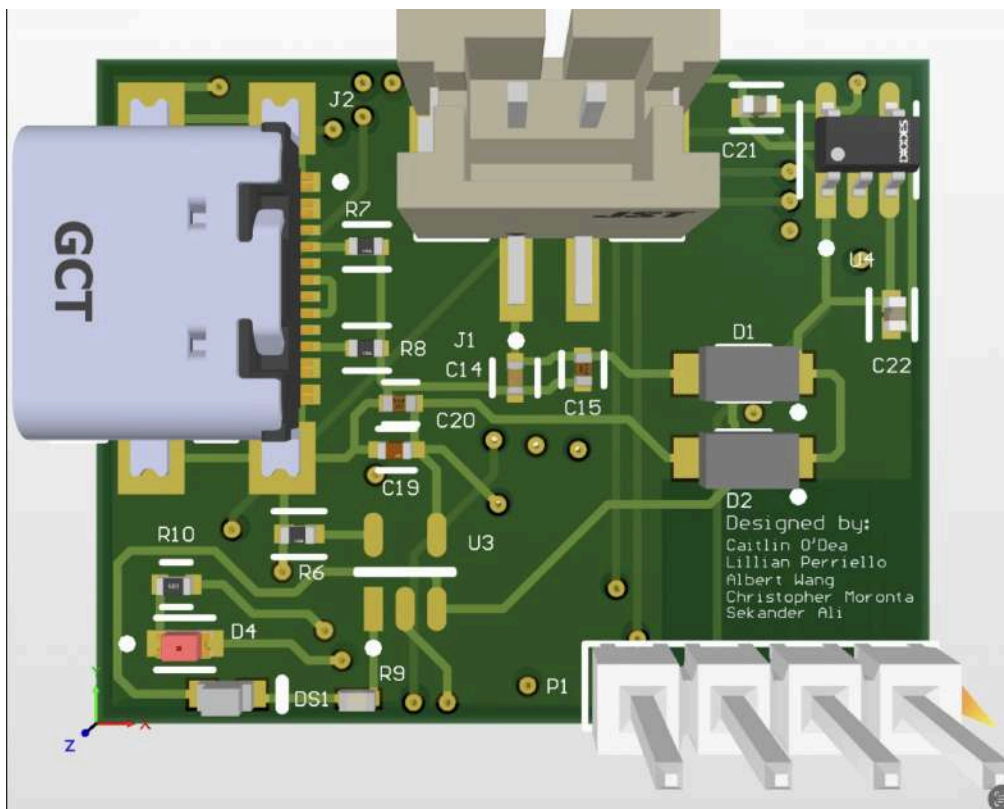
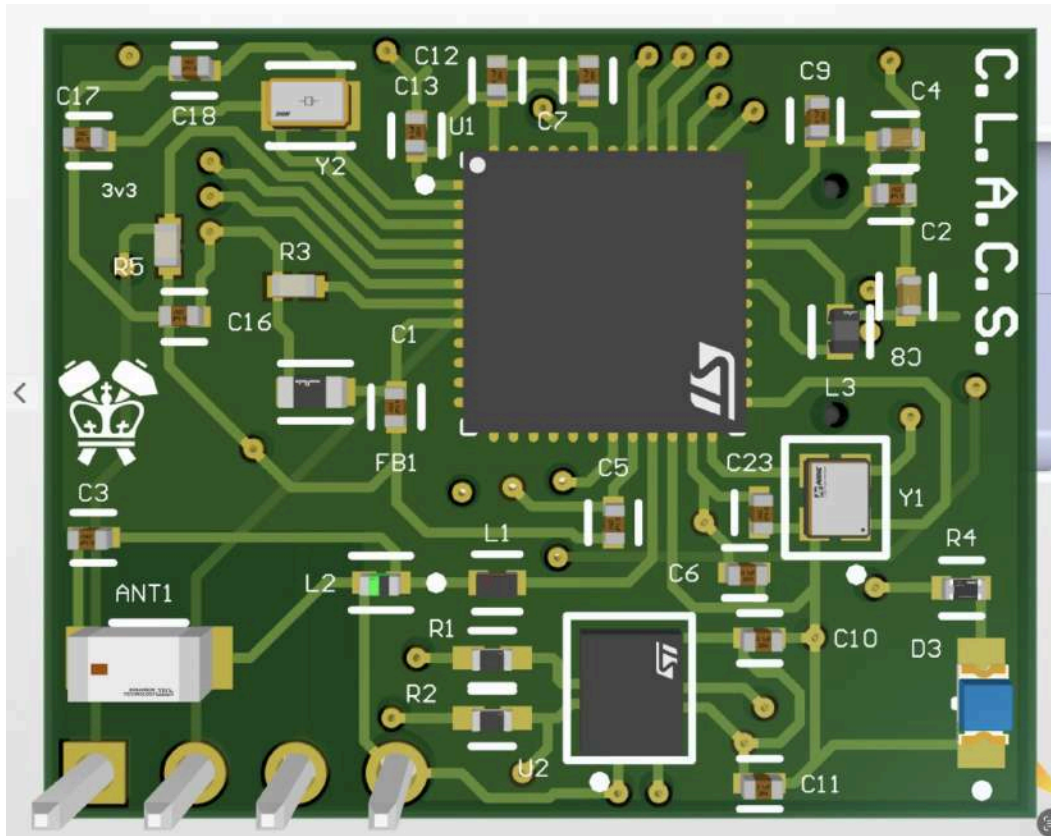




The group had many meetings in 1206 brainstorming, laying out the design in Altium, and collecting data. Here, we were discussing the design of the power circuit.

The following images show the PCB design in Altium:

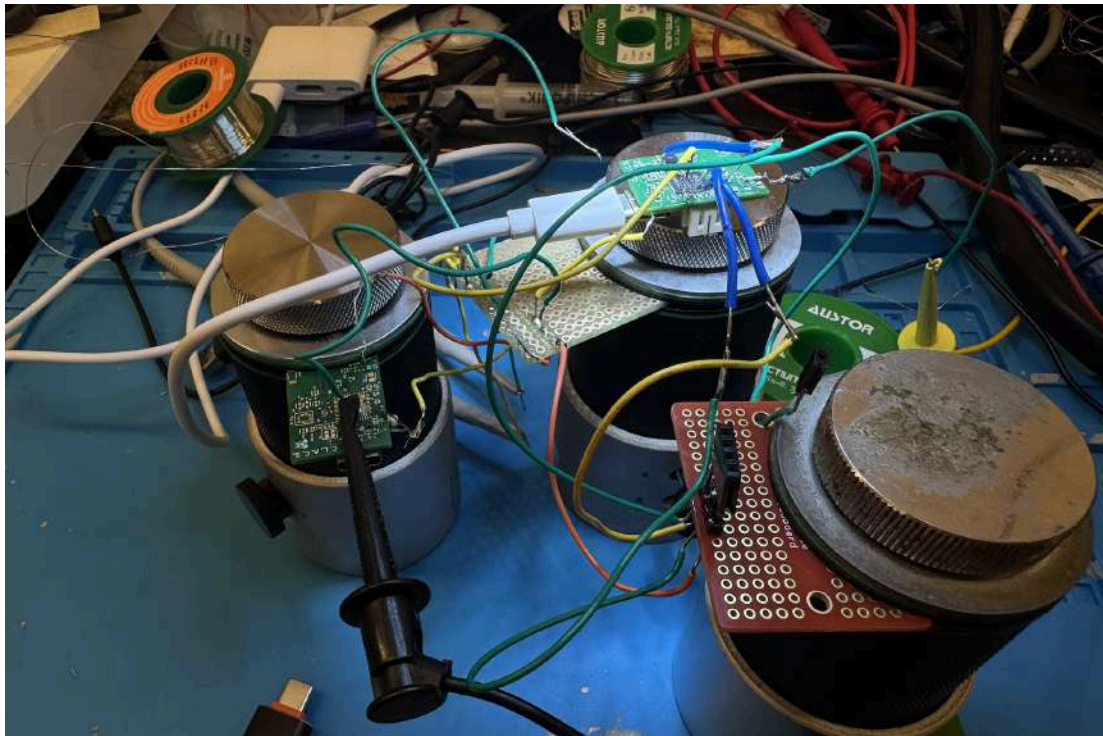


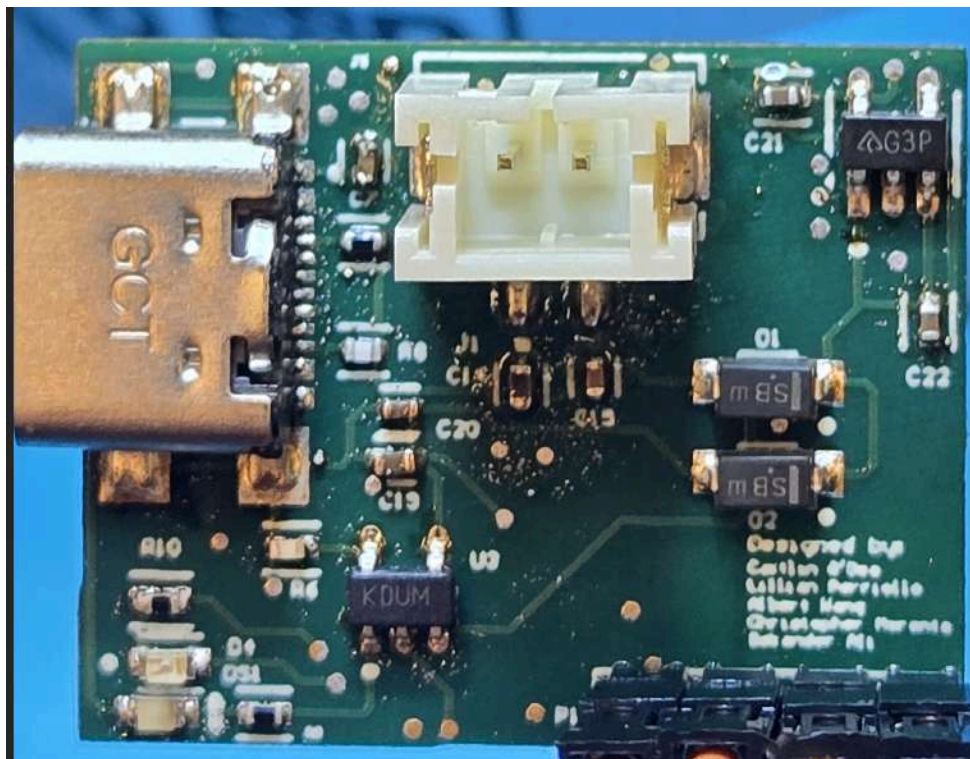


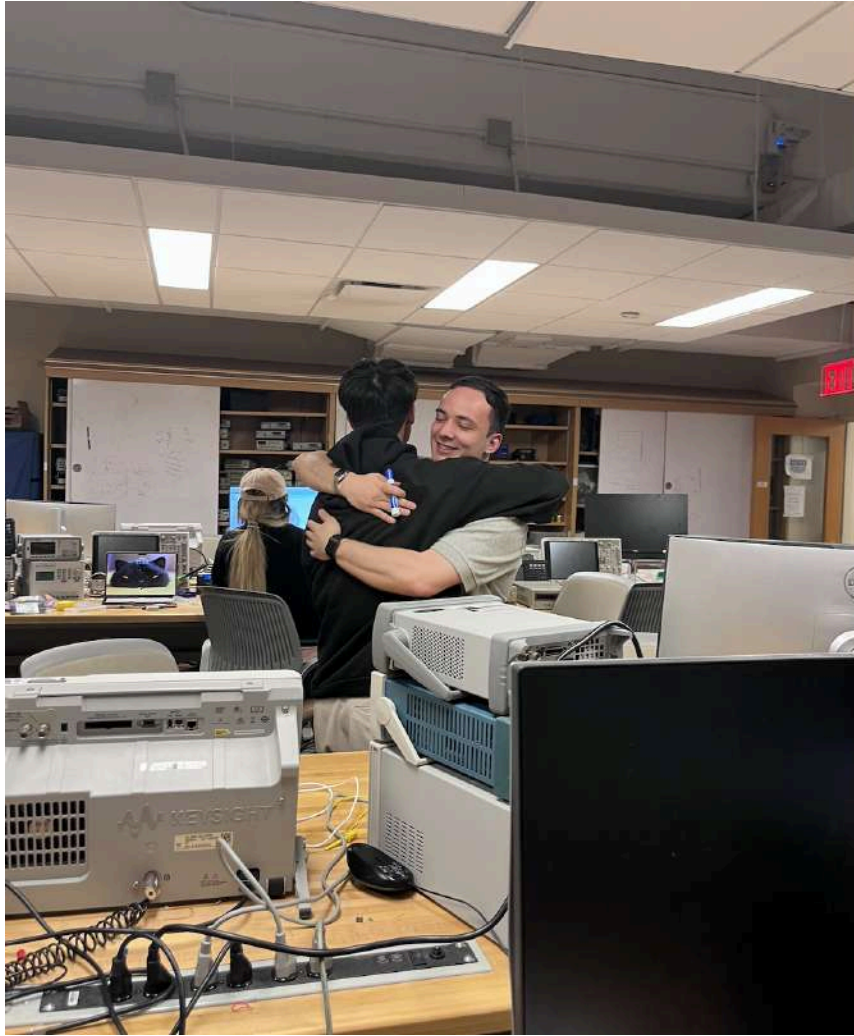
The team learned how to solder the components by hand. By the time of the Expo, we soldered four complete boards.



The image below shows the “Frankenstein Board” where we had multiple jumper wires in order to connect our board to our laptop, debug, and flash the board with our code.







Sekander and Albert were ecstatic when the board was finally working. All of the hard work was worth it!



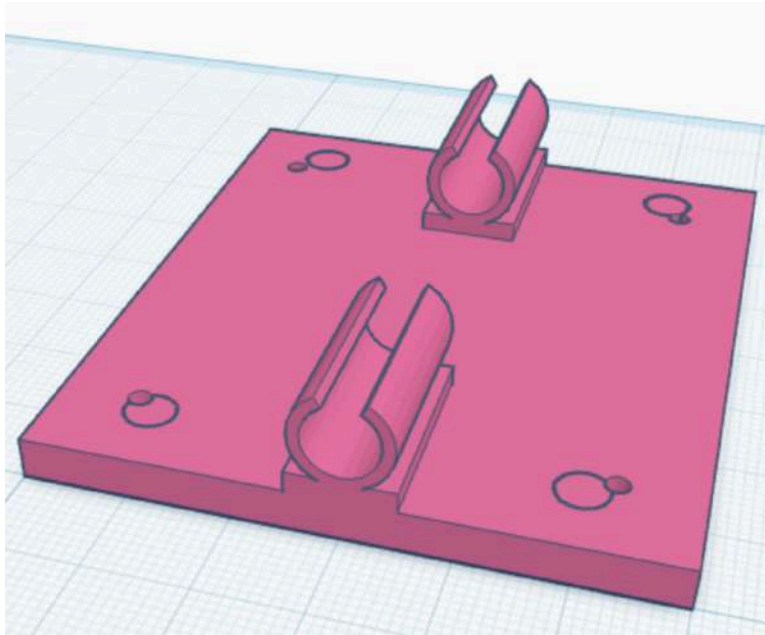
Expo Ready! Chris made a "PCB Museum" to showcase our progress with both PCB design and soldering. This museum also featured an extra copy of our 3D printed enclosure that holds the PCB and Li-Po battery pack. We also added a smaller section on the far left that allowed students to see components like resistors, capacitors, and the IMU without the board.

Master Commercial Parts List:

The following link shows the complete parts list for this project: [Master Commercial Parts List](#)

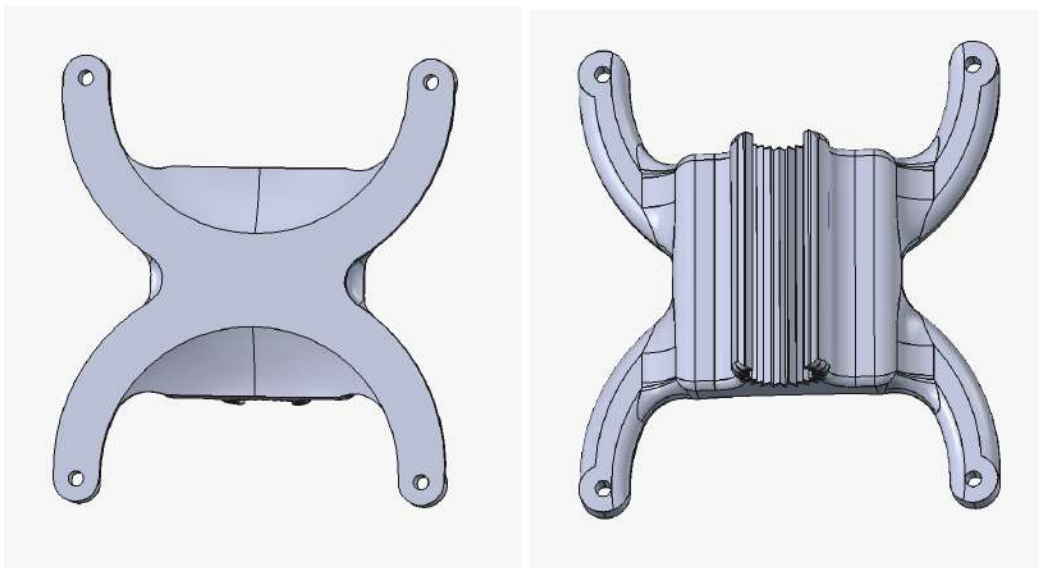
Makerspace Parts List:

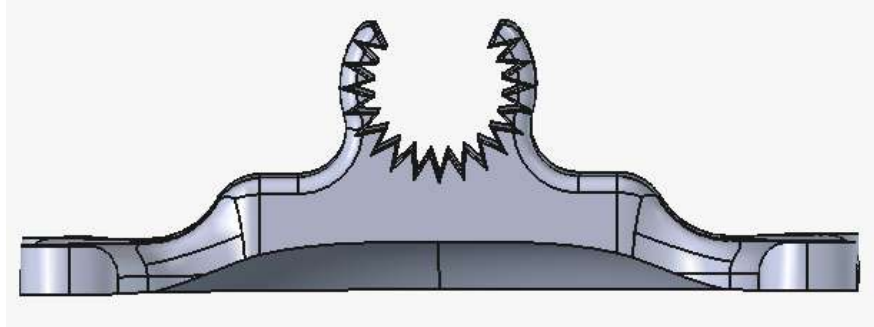
CAD .step and .stl files are attached within the files section of the CLACS Courseworks homepage.



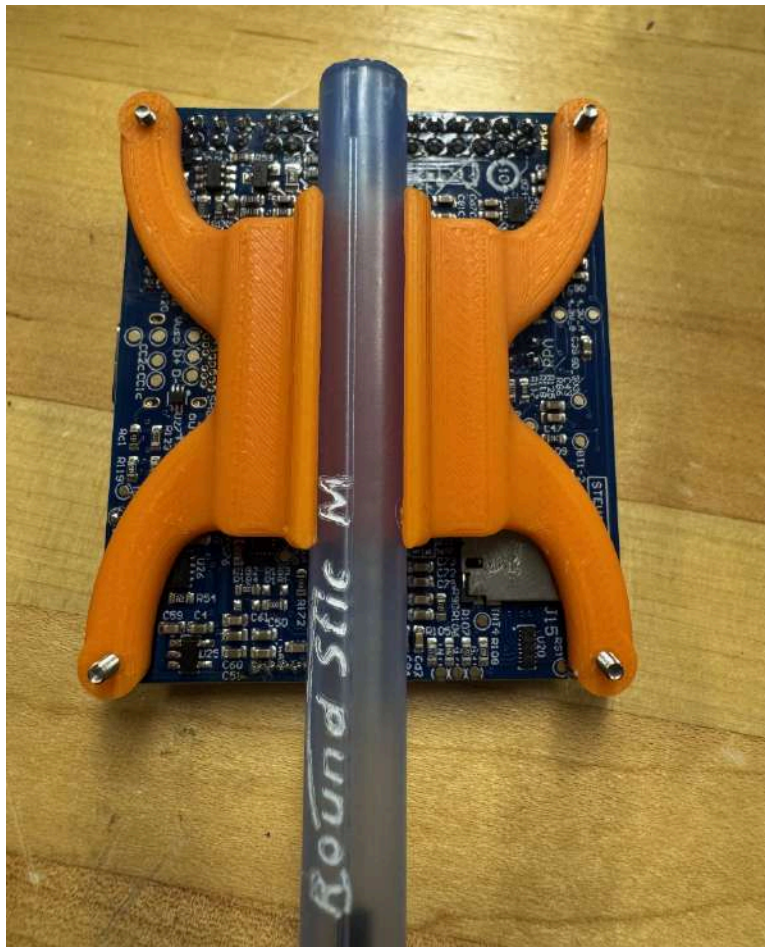
The image on the right shows the first successful prototype as a pen attachment. The clips on the back fit to standard No. 2 pencil and slim pens and the through holes in each corner allowed us to attach the test board onto the other side. This prototype was later redone since the clips were not strong enough to withstand continuous testing. This model was only used for the test board since this was made before we designed our PCB.

The images below show the modified version of the pen attachment for the test board. The test board was much larger than the PCB so we had to design something light and effective. We took the main ideas from the previous design (clip directly on the back and through holes for securement) and designed the following:

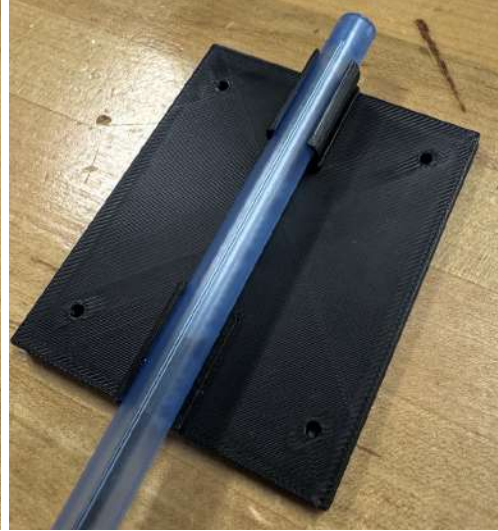
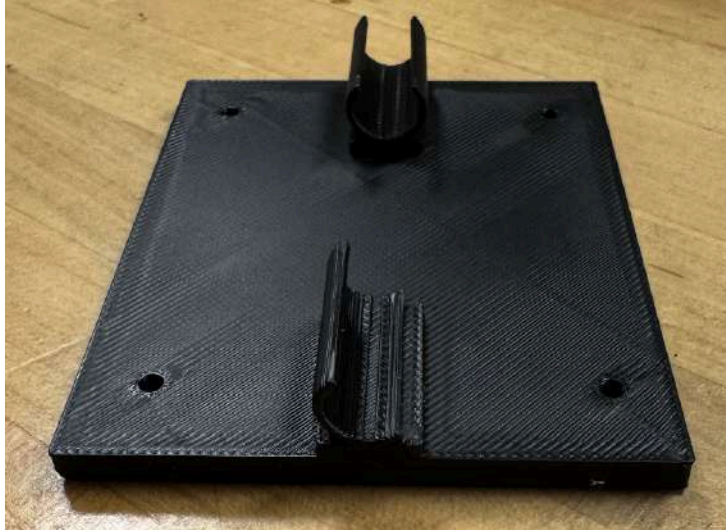




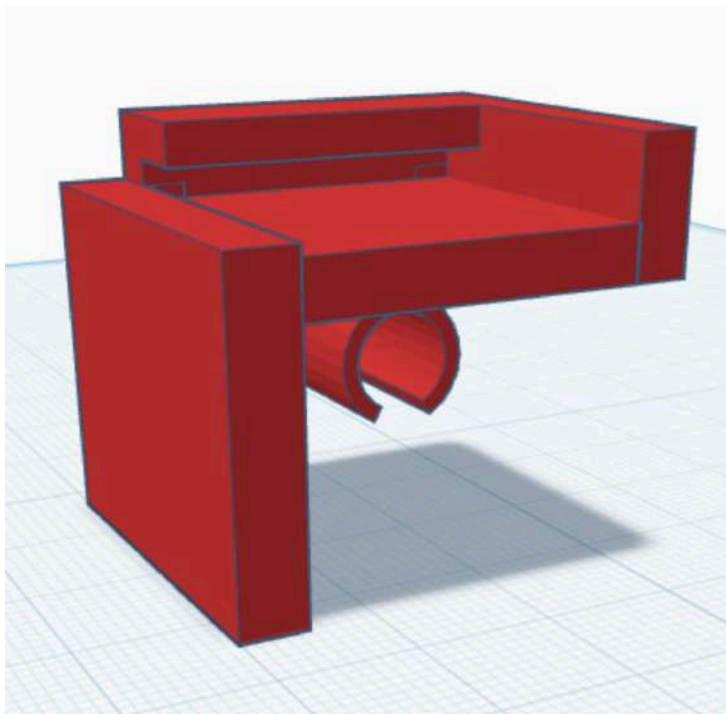
The clip design features many triangular ridges to create a strong attachment point that also allows for flexibility. In the image below you can see how the original design had a critical failure point on the clip itself.



The image on the left shows the final print for securing the test board. This was useful for the expo since it provided a stable attachment so other students and faculty could get an idea of how much our design has changed from the first prototype. We left this test board attached to one laptop to show how we were collecting data and the other pen attachment with the PCB was used for the live demonstration.



The bottom clip broke during initial data collection, prompting a complete redesign for improved durability.

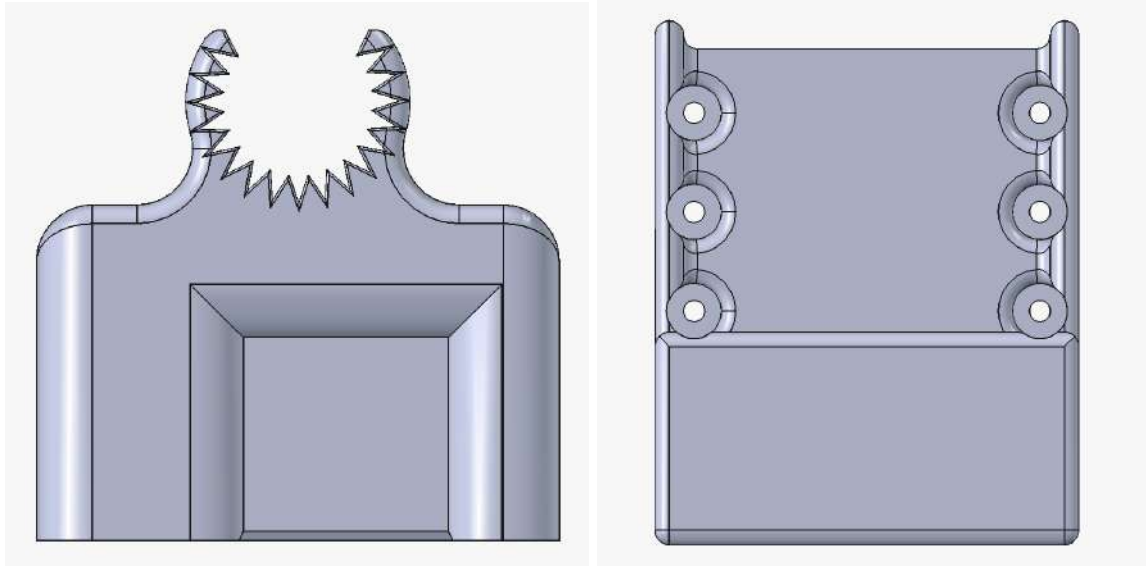


This was the first iteration of our pen attachment for the new PCB. We took the same clip from the original prototype but scaled the overall size down since the PCB was much smaller than our test board. The flat wall on the left side was for attaching the battery pack with adhesive and the flat surface on top was made so that the PCB could slide in vertically and be secured with the side walls while still having space for the larger components on the board that stuck out (USB-C and battery connector). Due to weight and structural concerns, this model was never printed. Instead of using STL files like the previous two models, we switched to STEP files for the final design. STL files chop up curved surfaces and create a rigid design that lacks flexibility and durability. The final design

using STEP files is shown below.

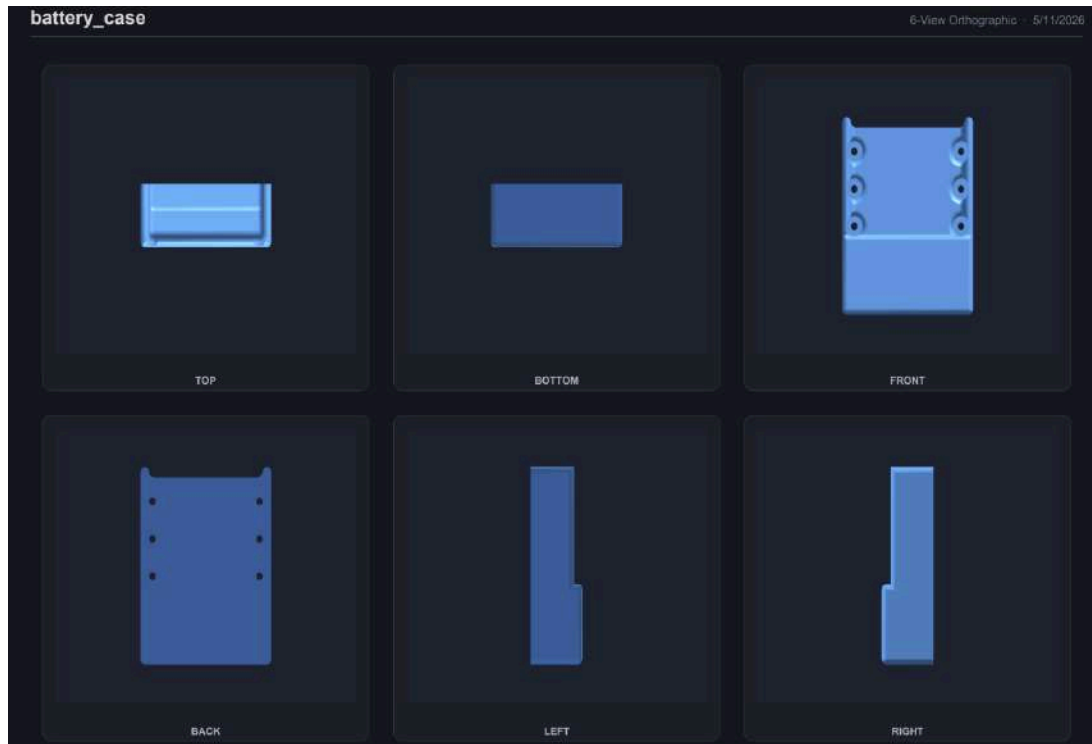
Final Design:

The images below show the pen attachment on the left and the battery case on the right. Both parts came together to create an enclosure that would hold both the PCB and battery while clipping onto the side of the pen.

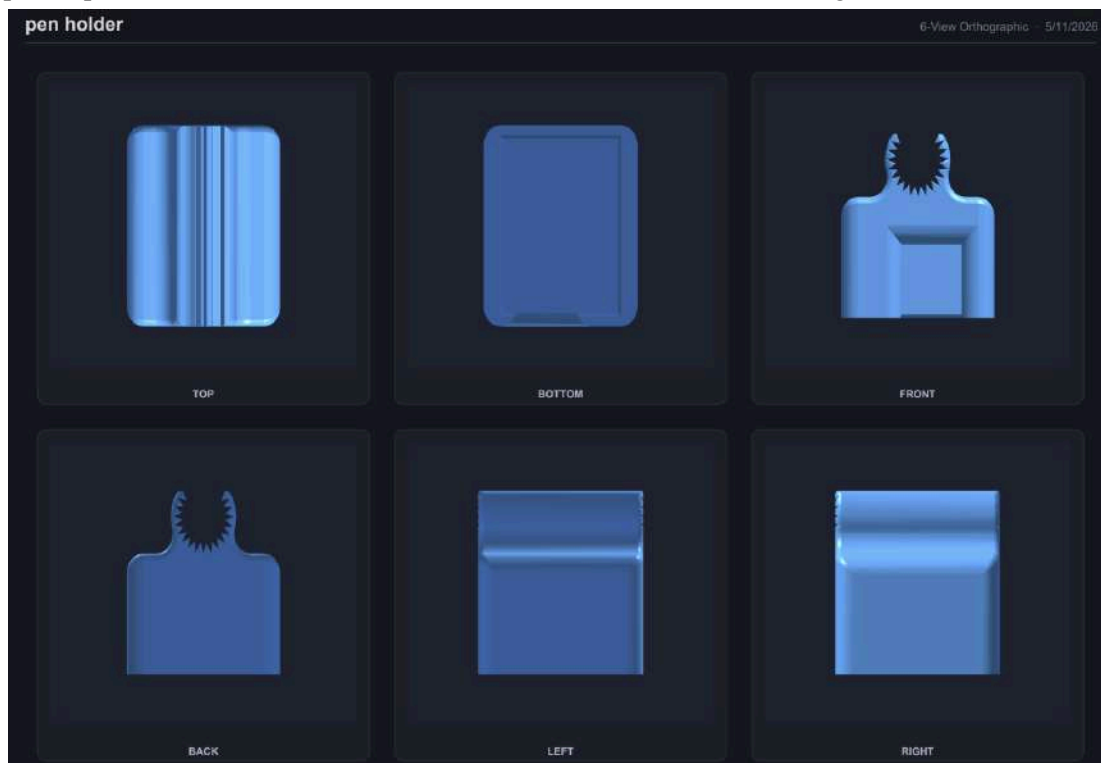


Below is the full orthographic view of the battery case. This was designed to be able to have the battery pack be easily removable and the screw holes allowed for attachment onto the pen clip.





The orthographic view of the pen attachment below shows the full dimensionality of this part. We started with the clip from the original pen attachment and then modified it to allow for stress and strain that the part experiences when the user attaches and detaches it from the writing utensil.



The small ridges within the clip itself allow the clip to be more flexible than before. This was a breaking point in our original design, so we made sure to modify this to allow the attachment to be ready for the live demo at the Senior Design Expo.

Engineering Standards:

1. **USB-C:** USB 3.1/USB Power Delivery USB-C provides high-speed data transfer and power delivery through a single connector. For CLACS, USB-C enables bidirectional communication with the host computer at up to 480 Mbps and allows the pen device to charge while operating. The connector's reversible design and durability make it ideal for portable applications.
2. **I2C:** Philips/NXP I2C Specification I2C is a two-wire serial bus protocol (SDA and SCL lines) designed for short-distance, low-speed communication between microcontroller and peripheral devices. In CLACS, I2C interfaces the STM32 microcontroller with the IMU sensor for real-time motion data acquisition. The protocol supports multiple devices on a single bus, reducing PCB routing complexity and power consumption compared to parallel interfaces.
3. **STM32 Microcontroller Architecture:** ARM Cortex-M4 32-bit RISC Architecture The STM32 microcontroller implements the ARM Cortex-M4 instruction set, ensuring compatibility with standard embedded development tools, libraries, and debugging protocols (SWD). This standard architecture provides deterministic real-time performance suitable for sensor data processing and is widely used in industrial and consumer applications requiring reliable embedded control.
4. **IMU Sensor Communication Protocol:** I2C Serial Interface (see I2C above) / Manufacturer Specifications The IMU (Inertial Measurement Unit) communicates motion data (3-axis acceleration, 3-axis angular velocity) to the microcontroller via standardized I2C protocol at 400 kHz. This allows synchronous data acquisition at configurable sample rates (typically 100–1000 Hz), enabling high-fidelity motion capture necessary for accurate handwriting reconstruction.
5. **PCB Design & Manufacturing Standards:** IPC-2221 (Generic Standard on Printed Board Design) / IPC-A-600 (Acceptability of Printed Boards) CLACS PCB design adheres to IPC industry standards for trace width, spacing, via placement, and signal integrity. Version 2 of the PCB achieved an ~84% size reduction from v1 while maintaining electrical performance and manufacturability. The design follows best practices for high-density layouts and proper thermal management for the STM32 and supporting components.
6. **Power & Battery Standards:** IEC 61960 (Secondary Batteries for Portable Applications) / USB Power Delivery Specification The battery system meets safety and performance standards for lithium-ion cells used in portable electronics. USB Power Delivery specification ensures safe power delivery over the USB-C interface at negotiated voltage and current levels, protecting both the device and host computer during operation.

7. **ML Model & Data Format:** IEEE 754 (Floating-Point Arithmetic) / Standardized Training Framework The neural network for letter reconstruction uses standard IEEE 754 floating-point representation for weight storage and inference computation. The model is trained and exported in an industry-standard format (e.g., TensorFlow/PyTorch), enabling reproducibility and transfer learning for future improvements.
8. **Digital Communication Protocol (Hardware-Computer Interface):** Custom Binary Protocol over USB-CDC (Communications Device Class) Data from the IMU is packetized into a custom binary protocol optimized for real-time transmission. USB-CDC (a USB standard subclass) allows the STM32 to appear as a virtual serial port on the host computer, enabling plug-and-play compatibility with Windows, macOS, and Linux without custom drivers.

PCB Specific Standards:

9. **IPC-2221C:** The generic design standard for printed boards that establishes minimum acceptable thresholds for trace width, spacing, via diameter, annular rings, and clearance, while allowing product-specific requirements to exceed those thresholds. This is what guides trace sizing to avoid overheating and ensure proper current flow.
10. **IPC-A-600 & IPC-6012:** IPC-A-600 is the visual acceptance standard showing what acceptable and unacceptable conditions look like through photographs and illustrations, while IPC-6012 specifies performance requirements. Together they ensure your manufactured PCBs are both visually acceptable and electrically reliable.
11. **IPC-J-STD-001:** The standard that specifies criteria for soldering components to PCBs, covering materials, methods, and inspection criteria to ensure the reliability and quality of soldered connections. This applies directly to your hand-soldered v2 PCB.

Constraints:

Power and Battery Constraints

Battery Capacity vs. Weight/Size Trade-off

- The single pen attachment must remain lightweight enough not to affect natural writing feel, fatigue the user's hand, or prove cumbersome to carry around
- Battery must provide sufficient runtime for typical use cases like class/work sessions
- **Design:** Battery pack slots directly into 3D-printed enclosure with dedicated connector to PCB
- Combined weight of pen attachment (PCB + housing + battery) kept minimal to preserve natural handwriting experience
- Battery capacity optimized for 2-4 hour continuous writing sessions

Power Budget

- Real-time Bluetooth transmission and IMU sensors continuously drain power
- Sensor sampling rates optimized to balance accuracy with power consumption
- Bluetooth module power management critical for extended runtime
- Device enters idle/low-power mode when pen not actively writing
- Battery life validated through extended use-case testing
- The model working for the Expo was hardwired to the computer for power and data transfer due to a last minute error with the IMU that required a jumper wire that prevented the attachment of the battery pack

Ergonomics and User Experience Constraints

Weight Distribution

- Single pen attachment must not significantly alter the pen's center of gravity
- Weight is critical; attachment position impacts moment of inertia, directly affecting writing feel
- 3D-printed housing designed to be as compact as possible while containing PCB and battery
- Combined weight target: <50g for entire pen attachment (PCB + housing + battery)
- Design Iteration: Original bottom clip design fractured during initial data collection; complete redesign improved durability and structural integrity

Attachment Dimensions

- Single collar-style attachment fits various pen/pencil diameters via adjustable clamping mechanism
- Attachment should not obstruct the user's view of the writing tip
- Should not extend too far from the pen body (target: <40mm length)
- 3D-printed enclosure provides weather-resistant housing for PCB and battery
- Redesigned attachment provides stronger grip without increasing encumbrance

Wireless Operation

- Battery-powered pen operates wirelessly via Bluetooth
- User can move freely without USB cable tethering
- Demo Limitation: Last-minute IMU power delivery issue required jumper wire workaround, preventing 3D housing installation; USB-C hardwiring used as temporary solution for expo presentation
- Final design restores intended Bluetooth wireless operation once IMU power issue resolved

Technical Performance Constraints

Accuracy Requirements

- Comparable accuracy to leading devices like Livescribe Symphony

- Target: $\pm 2\text{mm}$ positioning accuracy for handwriting reconstruction
- Must handle various handwriting styles and speeds
- Status: Achieved letter-level recognition (limited to 8 letters); word-level recognition in progress

Latency

- Real-time letter visualization requires minimal lag between physical writing and digital display
- Target maximum latency: 100ms (achievable with Bluetooth + local processing)
- Demo Note: USB-C hardwiring achieved $< 1\text{ms}$ latency for rapid ML model validation; final wireless design will trade minimal latency increase for mobility benefits

Sensor Limitations and Calibration

- IMU drift over time corrected through periodic calibration routines
- 6-axis accelerometer and gyroscope data processed in real-time via STM32
- Critical Issue Identified: IMU power delivery required jumper wire fix; final PCB design must route power cleanly to IMU to accommodate 3D enclosure
- ML model handles various writing angles and orientations during letter reconstruction
- Continuous data stream via Bluetooth allows for dynamic sensor fusion

Environmental and Usage Constraints

Writing Surface

- System designed to work on standard paper for classroom/office use
- No external base/reference frame required; pen-mounted sensors sufficient
- Weather-resistant 3D-printed housing protects electronics during portable use

Bluetooth Range and Interference

- Pen attachment must maintain reliable Bluetooth connection to host device (computer/tablet)
- Target range: 10+ meters for typical classroom/office environment
- Must function in environments with multiple Bluetooth devices (Wi-Fi networks, other wireless peripherals)
- Should handle brief signal interruptions without losing tracking data
- Bluetooth connection established at startup; graceful reconnection if interrupted

Portability

- Battery-powered design enables untethered writing experience
- User can move around room while writing without cable management concerns
- Pen attachment remains comfortable during extended writing sessions

Manufacturing and Material Constraints

Component Availability

- STM32 microcontroller: readily available, cost-effective
- IMU sensor (6-axis): standard component, widely sourced
- Bluetooth module: standard low-energy module for wireless communication
- Lithium battery: small-form-factor (coin cell or similar) for pen attachment
- 3D-printed housing: PLA filament printed on standard Bambu Lab FDM printer

Durability

- Primary Challenge: Bottom collar clip experienced failure during early testing
- Solution: Complete redesign implemented for v2, achieving robust attachment mechanism
- Redesigned clip withstands repeated attachment/detachment cycles
- Components survive drops and normal handling
- 3D-printed enclosure provides structural support and environmental protection
- Manufacturing Issue: PCB v2 requires clean power routing to IMU; jumper wire workaround must be eliminated in final design for housing to fit properly
- Achievement: ~84% PCB size reduction from v1 to v2 while maintaining structural integrity and fitting within 3D enclosure

PCB Design Standards

- IPC-2221C compliance for trace width, spacing, and thermal management
- IPC-J-STD-001 hand-soldering standards for reliable component attachment
- IPC-A-600 visual acceptability standards for manufactured PCBs
- Critical: Final PCB iteration must resolve IMU power delivery to enable complete 3D housing assembly

Safety Constraints:

Battery Safety

- Lithium battery (coin cell format) enclosed within 3D-printed housing
- Protection against puncture, crushing, or short circuit
- Battery holder provides secure seating with proper polarity protection
- Temperature monitoring during extended use sessions
- Battery connector provides safe, keyed connection to PCB

Electromagnetic Compatibility

- Bluetooth module operates within ISM 2.4 GHz band (IEEE 802.15.1)
- Must coexist with Wi-Fi and other 2.4 GHz devices in typical environments

- PCB layout minimizes RF interference with other electronics
- Antenna placement optimized for reliable Bluetooth signal within pen attachment housing

Software and Data Constraints

Processing Requirements

- Real-time letter reconstruction on host computer (Python/TensorFlow)
- IMU data streamed wirelessly via Bluetooth to host device
- STM32 handles sensor buffering and Bluetooth packet transmission
- Algorithm filters noise and distinguishes writing strokes in post-processing on host side
- Bluetooth connection established at startup; automatic pairing for seamless user experience

Data Transmission

- Wireless Bluetooth link eliminates need for USB cable during normal operation
- Continuous streaming of IMU data (6-axis) at configurable sample rates
- Bluetooth Low Energy (BLE) considered for power efficiency
- Connection drops handled gracefully; data buffered locally if interruption occurs
- No custom drivers required on host (Bluetooth pairing handled by OS)

Brag Sheet:

Attached separately by each group member.